Class: TE **Subject: AISC** **Sem-V**

Experiment No: 5

Course Outcome: LO2 **Blooms Level: L3**

Aim: Implement Predicate Logic using Library Management System

Abstract: This experiment demonstrates the implementation of Predicate Logic using a simple Library Management System. Facts such as student details, book issue status, membership, and fine payment are represented as predicates. Based on predefined inference rules, the program derives conclusions such as borrower eligibility, fine applicability, book availability, and borrowing permission. The experiment helps students understand the concepts of Knowledge Representation, Predicate Logic, and Rule-Based Inference in Artificial Intelligence.

Sample Input and Output:

Enter Student Name: Rahul
Enter Book Name: Artificial Intelligence

Is the book issued? (yes/no): yes
Enter Number of Late Days: 5
Is the book reserved? (yes/no): yes
Is Membership Active? (yes/no): yes
Has Fine been Paid? (yes/no): no

========== FACTS ==========

Student(Rahul)
Issued(Rahul, Artificial Intelligence)
DaysLate(Rahul,5)
Reserved(Artificial Intelligence)
MembershipActive(Rahul)

========== INFERENCE ==========

Borrower(Rahul)
Fine(Rahul)
BookUnavailable(Artificial Intelligence)
Rahul must pay the fine before borrowing again.

Class: TE **Subject: AISC** **Sem-V**

Theory:

Predicate Logic, also known as **First Order Logic (FOL)**, is a knowledge representation technique used in Artificial Intelligence to represent **objects, facts, relationships, and rules**. Unlike propositional logic, predicate logic can represent relationships among objects using **predicates**.

A predicate consists of:

Predicate(Object)

Example:

Student(Rahul)
Issued(Rahul, AI)
MembershipActive(Rahul)

Inference is performed using **IF–THEN rules**.

Example:

IF DaysLate(Student) > 0
THEN Fine(Student)

In this experiment, facts entered by the user form the **Knowledge Base**, and the program applies inference rules to derive new conclusions.

Algorithm

1. Start the program.
2. Accept student details and library information from the user.
3. Store the entered information as facts (predicates).
4. Display the generated facts.
5. Apply the inference rules:
   - If the book is issued, infer **Borrower(Student)**.
   - If the number of late days is greater than zero, infer **Fine(Student)**.
   - If the book is reserved, infer **BookUnavailable(Book)**.
   - If membership is active and the fine is paid, infer **CanBorrowAgain(Student)**.
6. Display all inferred conclusions.
7. Stop the program.

Class: TE **Subject: AISC** **Sem-V**

Code:

https://colab.research.google.com/drive/10vQ5zDQv2aRdd9SuOPZLCGKg-1D_NVDc?usp=sharing

Conclusion:
The experiment successfully demonstrates the implementation of Predicate Logic for representing knowledge and performing rule-based inference. Facts entered by the user are treated as a knowledge base, and logical rules are applied to derive meaningful conclusions. This experiment provides a practical understanding of Knowledge Representation and Inference Mechanisms used in Artificial Intelligence.

Exercise 1:

Extend the Library Management System using Predicate Logic
Modify the program to include the following predicates:

BooksIssued(Student, NumberOfBooks)
MembershipExpired(Student)

Implement the following inference rules:

If BooksIssued(Student) > 3, infer BorrowingLimitExceeded(Student).
If MembershipExpired(Student), infer BorrowingDenied(Student).

Exercise 2:

Design a Predicate Logic-based Hospital Management System. Develop a Python program using Predicate Logic for a Hospital Management System. Represent the following predicates:

Patient(Name)
Fever(Name)
Cough(Name)
Insurance(Name)
Age(Name)

Implement the following inference rules:

If Fever and Cough, infer Flu(Patient).
If Insurance, infer CashlessTreatment(Patient).
If Age > 60, infer SeniorCitizen(Patient).
If SeniorCitizen and Insurance, infer PriorityTreatment(Patient).

Display all generated facts and inferred conclusions. This exercise reinforces the concepts of Predicate Logic, Knowledge Representation, and Rule-Based Reasoning.